CCPN

CcpNmr Analysis Version 3

Side-chain Assignment Tutorial

Introduction

This tutorial is designed to guide you through the assignment of the aliphatic side chain resonances of a protein using Ccpnmr AnalysisAssign Version 3.1. It is not intended to teach any theoretical aspects of NMR. A wide range of NMR experiments can be used to do such an assignment and typically each laboratory has its own "favourite" set of experiments to record. The experiments and method shown here are therefore simply "a" method, rather than "the" method. But it should be easy to transfer this to other experiments you may have available for your own protein.

If you are new to NMR, we recommend that you have a look at our Understanding NEF Atom Names video which will help you understand the way you should name your side-chain atoms.

This tutorial uses spectra recorded on the CW domain of ASHH2 bound to a histone tail peptide. We are grateful to Dr Olena Dobrovolska at the University of Bergen for providing these. For more information on this system see Dobrovolska et al. 2020, FEBS Journal (https://doi.org/10.1111/febs.15256). You can download the tutorial data from the tutorials page of the CCPN website (https://ccpn.ac.uk/support/tutorials/).

Please note that the images shown are only representative and you may encounter minor differences in your setup.

Contents:

1. Assignment strategy and reference material
2. Assigning short side chains
3. Assigning long side chains

Start CcpNmr Analysis V3

Linux anad Mac users by using the terminal command: *bin/assign*

Windows users by double-clicking on the *assign.bat* file

1 Assignment Strategy

(Explanation only)

Short side-chain amino acids

We will start by assigning those amino acids that only have alpha and beta carbons/hydrogens in the aliphatic region. These residues are:

Glycine (GLY)
Alanine (ALA)
Serine (SER)
Cysteine (CYS)
Aspartic Acid (ASP)
Asparagine (ASN)
Phenylalanine (PHE)
Tyrosine (TYR)
Tryptophane (TRP)
Histidine (HIS)

For these residues, the CA and CB assignments are already available from the Backbone assignment. Only the HA and HB assignments are still required and can be obtained from the HBHACONH spectrum.

Long side-chain amino acids

The remaining side-chains have additional aliphatic atoms which need to be assigned:

Threonine (THR)
Glutamic Acid (GLU)
Glutamine (GLN)
Valine (VAL)
Leucine (LEU)
Isoleucine (ILE)
Arginine (ARG)
Lysine (LYS)
Methionine (MET)
Proline (PRO)

Assigning these amino acids requires additional spectra. We will be using the HBHACONH to assign the HA and HB atoms, the CCCONH spectrum to assign all the remaining side-chain carbon atoms and the 13C-HSQC and hCCH-TOCSY spectra for the remaining side-chain hydrogen atom assignments.
Note that some side chains such as Arg and Lys are often highly dynamic and their resonances may overlap significantly, making assignment difficult. Met methyl groups can also be difficult to assign, as they are separated from the remaining sidechain by an NMR-silent sulphur atom.

Aromatic side-chains

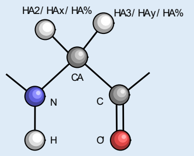
The aromatic side-chain elements of

Phenylalanine (PHE)
Tyrosine (TYR)
Tryptophane (TRP)
Histidine (HIS)

need to assigned separately using a further set of experiments, not shown here.

Natural Amino Acids

Amino Acids with NEF Atom Names (for referernce)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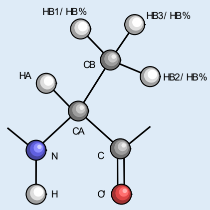
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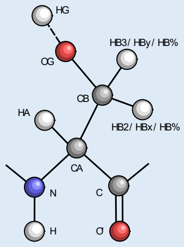
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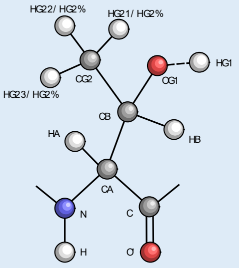
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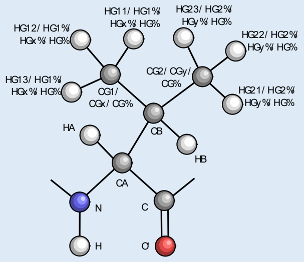
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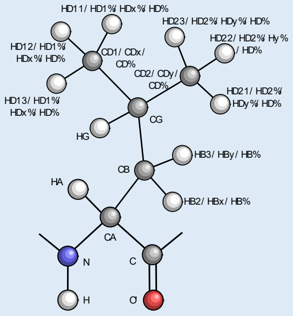
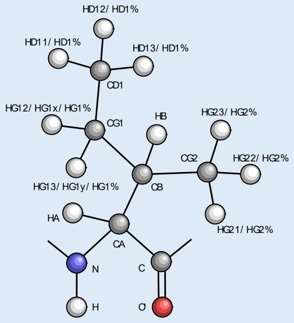
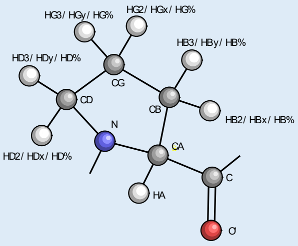
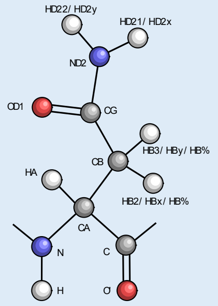
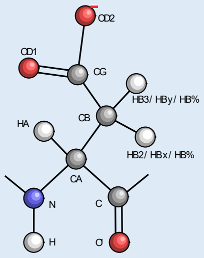
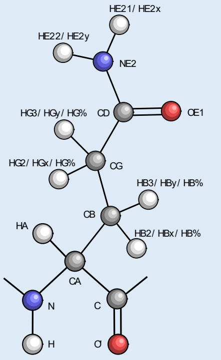

Leucine

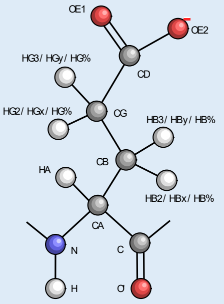

Isoleucine


Proline


Asparagine


Aspartate


Glutamine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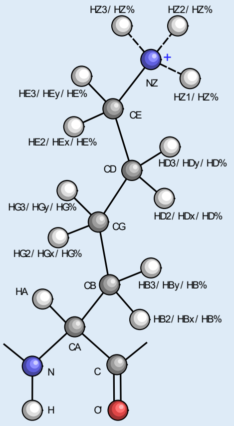
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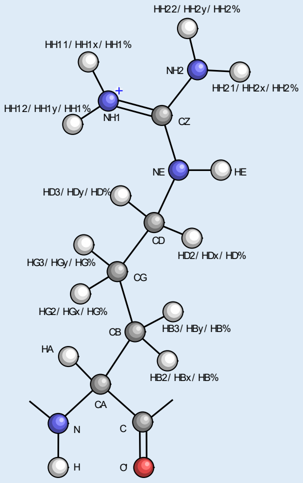
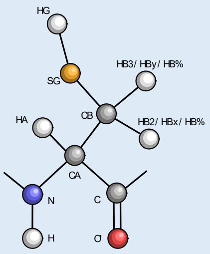
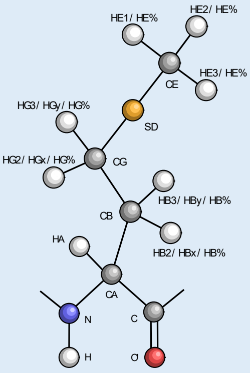
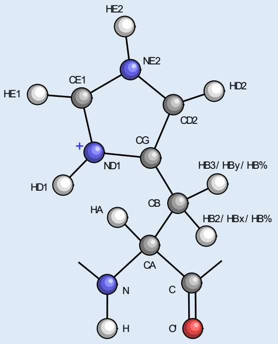
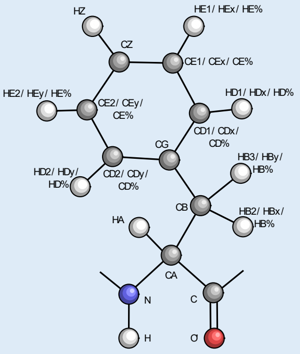
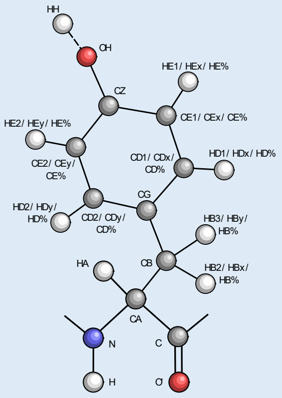

Arginine

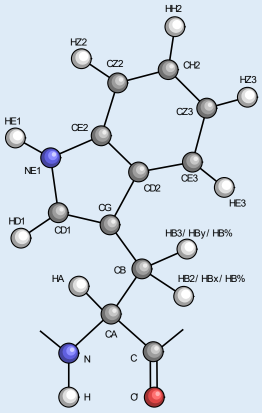

Cysteine


Methionine


Histidine

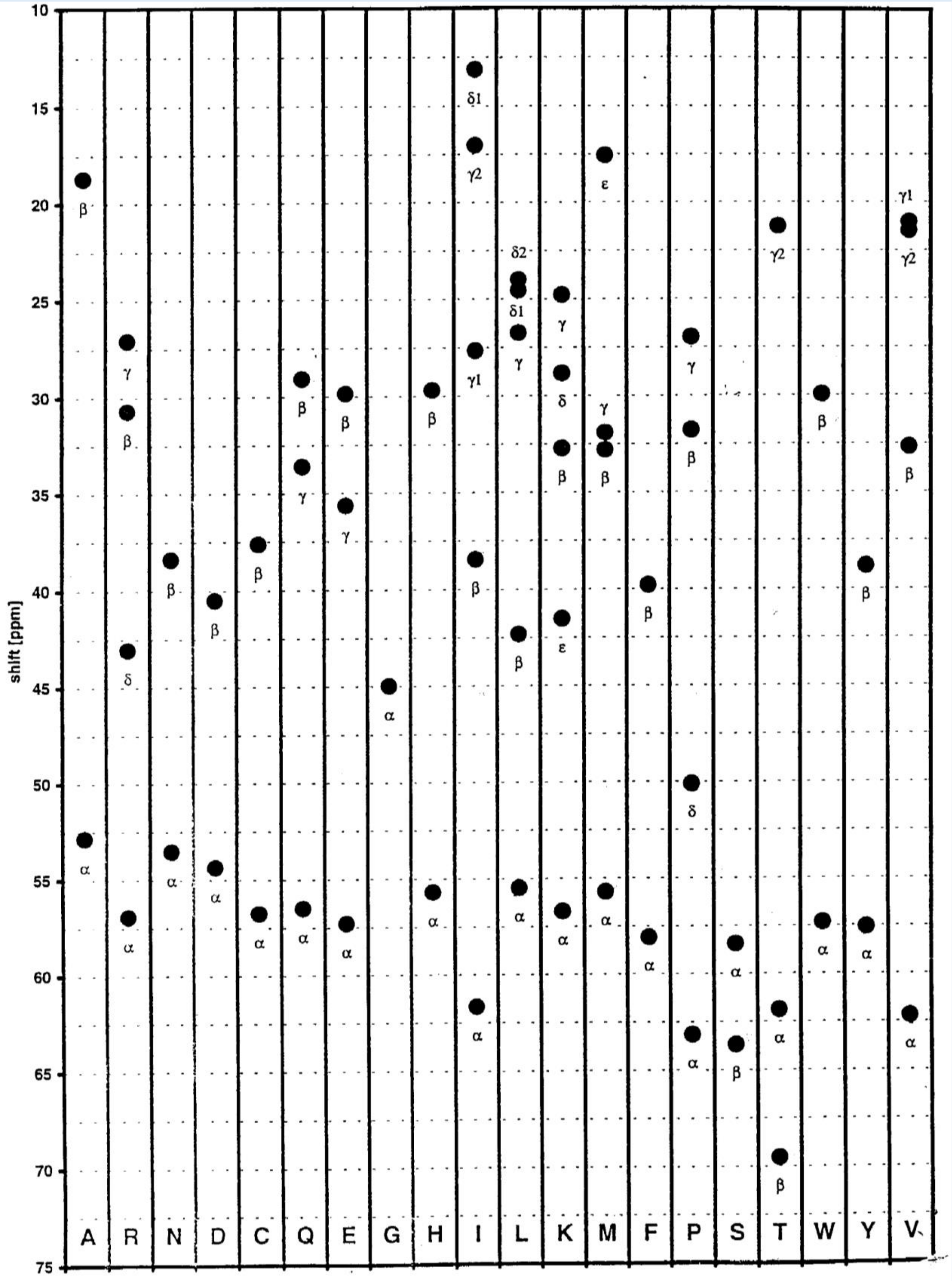

Phenylalanine


Tyrosine


Tryptophan

Standard Chemical Shift Values

Carbon chemical shifts for the 20 natural amino acids (for referernce)



1 Basic Program Usage

Getting started, basic operations

Sidebar

All data contained in a project, such as spectra and peak lists are located in the sidebar. **Double-clicking** on an item will open the properties popup.

Display

A display can contain multiple overlaid spectra. To show/hide a single spectrum, click on its toolbar button. If you close a display, you can open a spectrum by **dragging and dropping** it into the drop area from the sidebar or **right-clicking** on a sidebar item and selecting **Open as module**.

Mouse

- Pan -> **Left-drag** in display
- Zoom in/out -> **Scroll wheel** in display
- Context menu -> **Right-click**
- Select a peak -> **Left-click** on a peak symbol "X"
- Move a peak -> select first, then **right-click and drag**

Shortcuts

The program uses several shortcuts, for example **CL** for copying a peak list. You will need to press the first letter on your keyboard e.g. **C**, followed by the second letter, e.g. **L** (case insensitive). Press **Esc** to cancel the first letter.

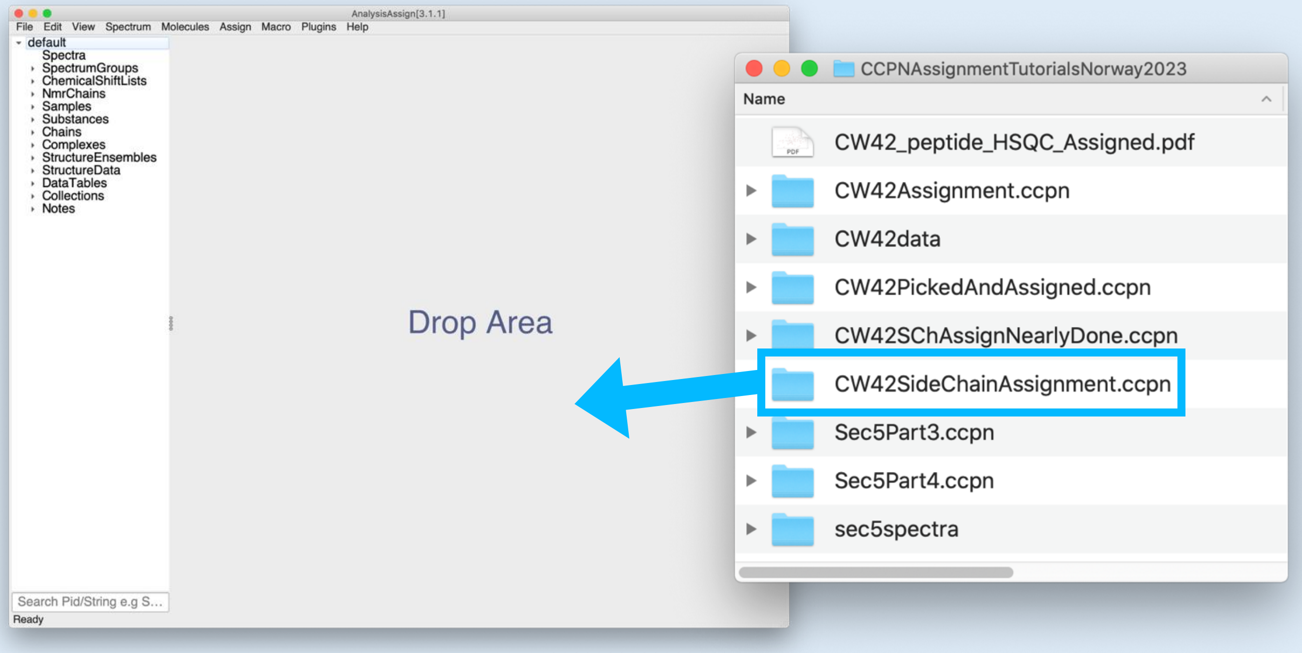
Fore more commands and operations:

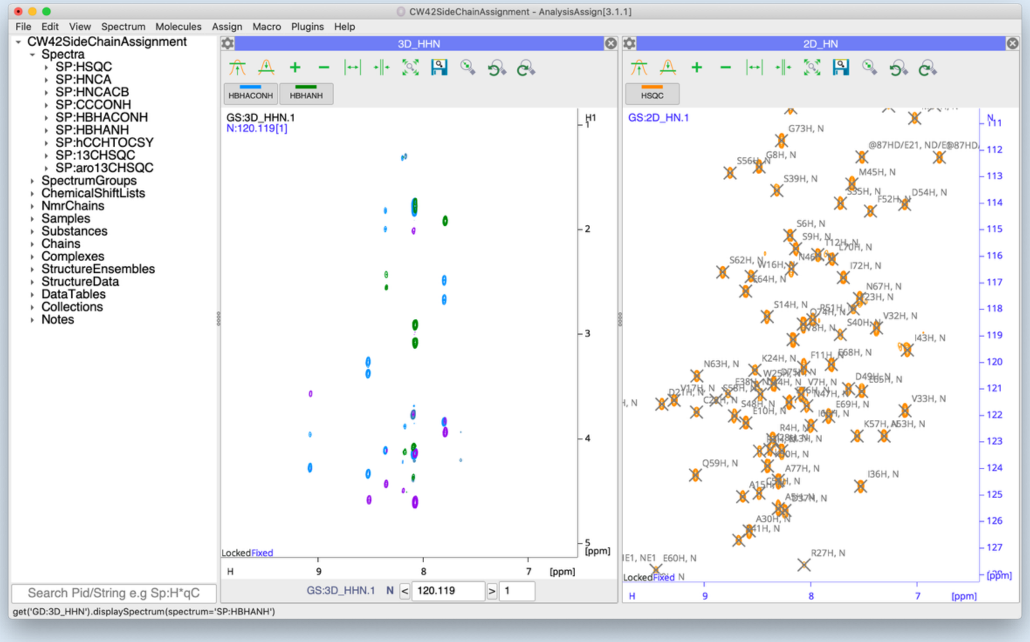
Main Menu -> *Help -> Tutorial (Beginners) or Show Shortcuts*

2 Assigning Short Side Chains

CW42SideChainAssignment

Open the project CW42SideChainAssignment.ccpn





2A Open Project

- Drag & drop the **CW42SidChainAssignment.ccpn** folder into the sidebar or drop area.

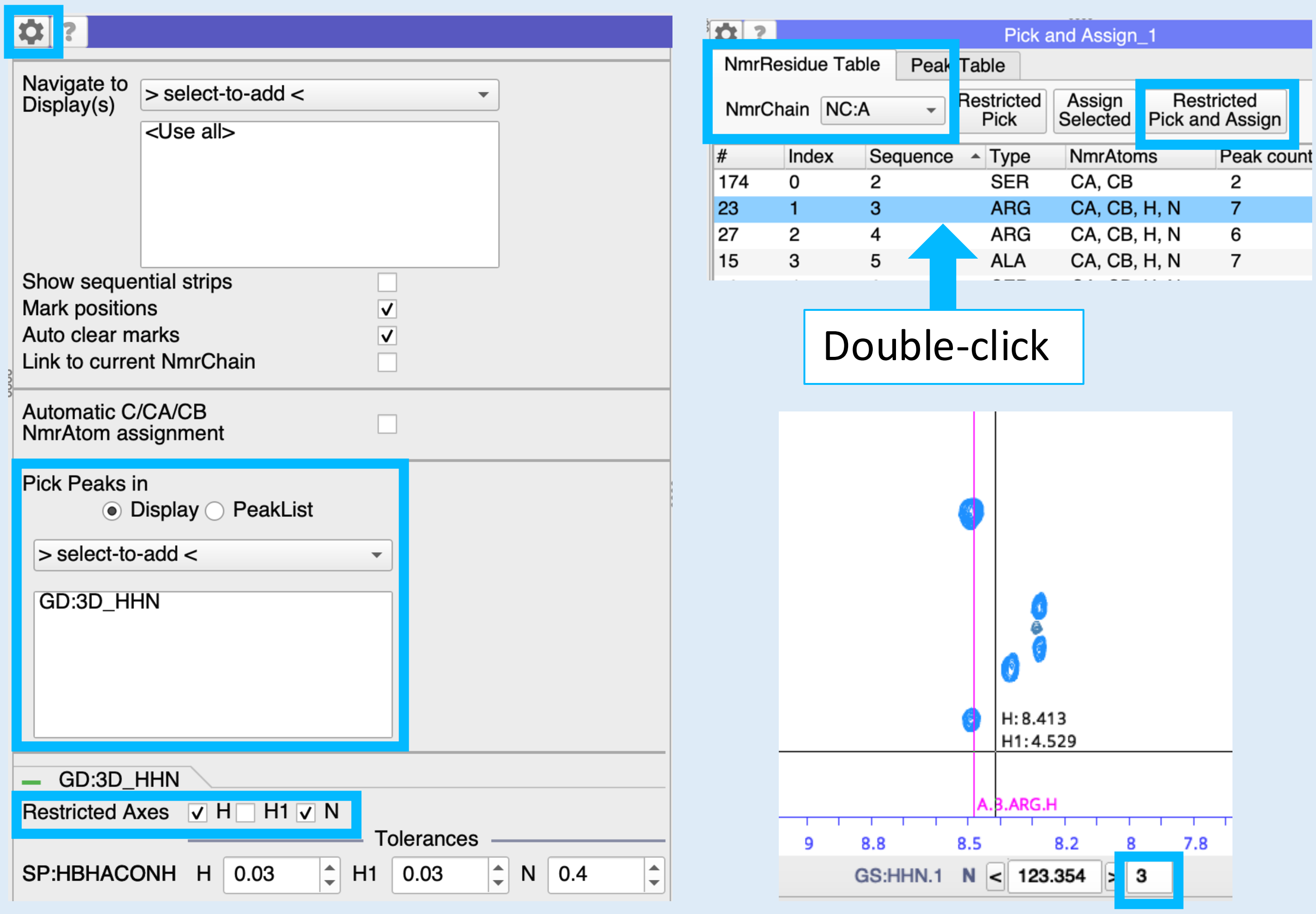
You will see two spectra, displayed in two Spectrum Displays as:

- HSQC (orange, 2D_HN)
- HBHACONH (blue, 3D_HHN)

with additional spectra in the sidebar.

Assigning Short Side Chains

CW42SideChainAssignment



2B Pick and Assign HBHACONH

Open the **Pick and Assign** module:

- Go to **Main Menu** ⟶ **Assign** ⟶ **Pick and Assign**, or shortcut **PA**.
- Open the Settings (gearbox icon
    - Pick Peaks in Display **3D_HHN**
      (if required, remove **<Use All>** by **right-clicking** and selecting **Remove**)
    - Make sure that only **H** and **N** are selected as **Restricted Axes**

In the main part of the **Pick and Assign** module:

- Select the NmrChain: **NC:A**.
- **Click** on the **Sequence** column heading to sort the table by sequence number.
- **Double-click** on a row in the table, e.g. the row for **3 ARG**:

  This will cause the **2D_HN** and **3D_HHN** SpectrumDisplays to focus on the position of the NH resonances of Arg 3 and mark their chemical shifts.

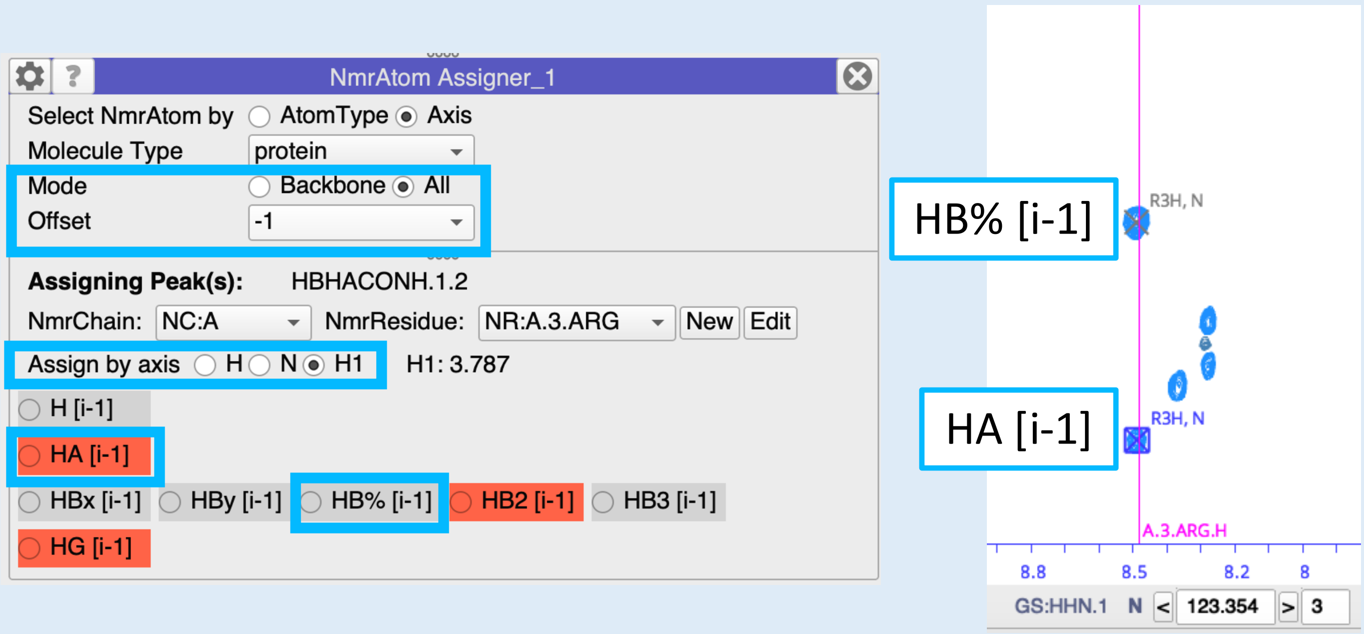
In the **3D_HHN** Spectrum Display:

- increase the plane count from **1** to **3**
- set the contour level high enough not to show negative or noise peaks.

In the **Pick and Assign** module:

- Click on **Restricted Pick and Assign** to pick the peaks in the **HBHACONH** spectrum and assign the H and N dimensions.

2 Assigning Short Side Chains



2c Assigning HA/HB NmrAtoms

Open the NmrAtom Assigner with

- Go to **Main Menu ⟶ Assign ⟶ NmrAtom Assigner**, or shortcut **AN**.

Open the Settings (gearbox icon and adjust the settings:

- Set **Mode** to **All**
- Set **Offset** to **-1**

In the main module

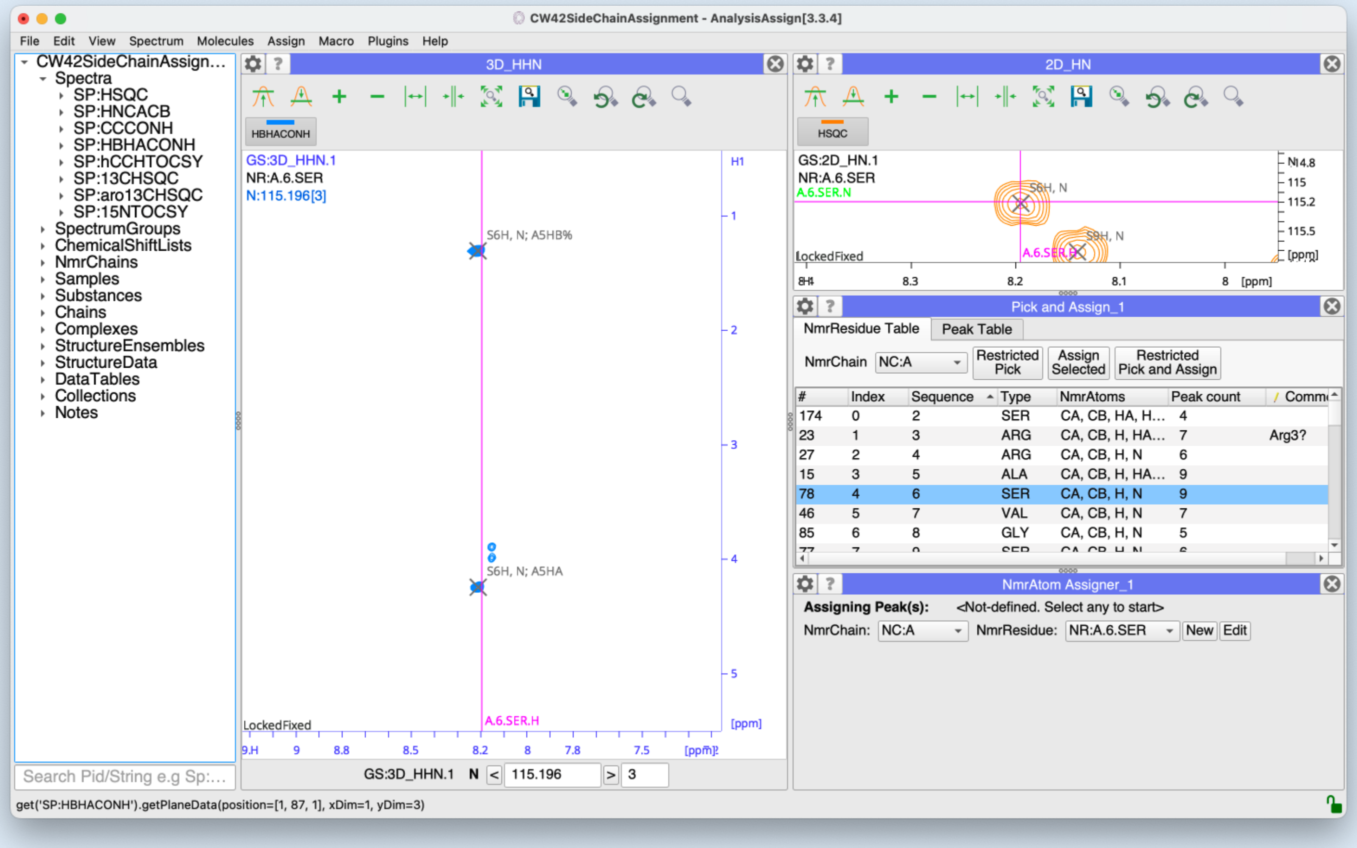
- Make sure that **Assign by axis** is to set to **H1**

- Select the lower peak and **click** on the **HA [i-1]** button
- Select the upper peak and **click** on the **HB% [i-1]** button

Note that if you have two HB peaks, you need to assign them to **HBx [i-1]** and **HBy [i-1]** (doesn't matter which way round). **HB2**/**HB3** should only be used in the unlikely situation that you are making stereospecific assignments.

The assignment options shown in the lower part of the NmrAtom Assigner will adjust depending on the i-1 amino acid type. For each amino acid type you will only be offered its possible atom names.

See secton 1 for a drawings of all amino acids and their NEF atom names. And have a look at our Understanding NEF Atom Naming video at https://www.ccpn.ac.uk/manual/v3/NEFAtomNames.html to understand the way in which the % and x/y nomenclature is used.



2D Further assignments

Arrange your modules as shown above and repeat the procedure outlined in sections 2B and 2C for further amino acids with short aliphatic sidechains. These are:

Glycine (GLY)

Alanine (ALA)

Serine (SER)

Cysteine (CYS)

Aspartic Acid (ASP)

Asparagine (ASN)

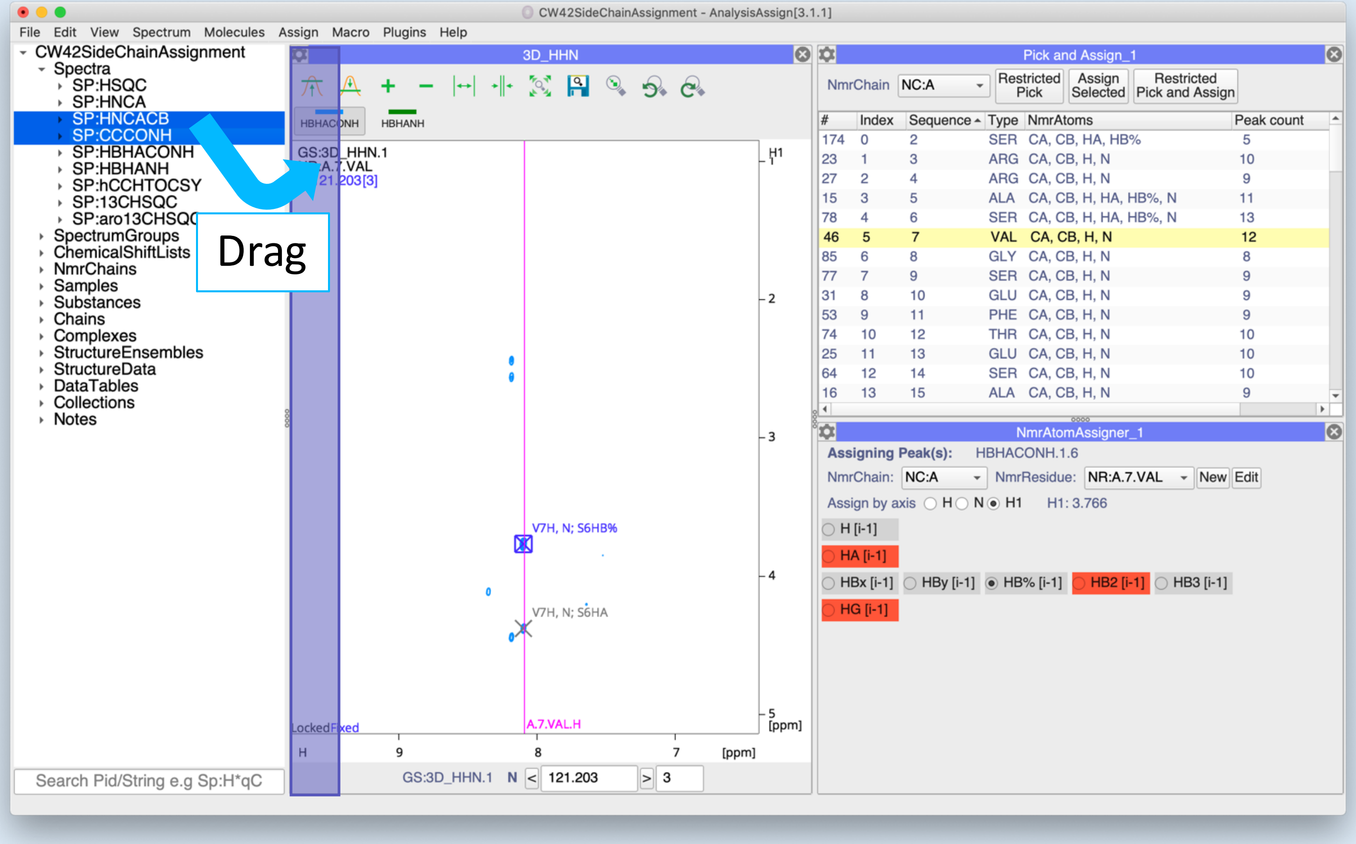
Phenylalanine (PHE)

Tyrosine (TYR)

Tryptophane (TRP)

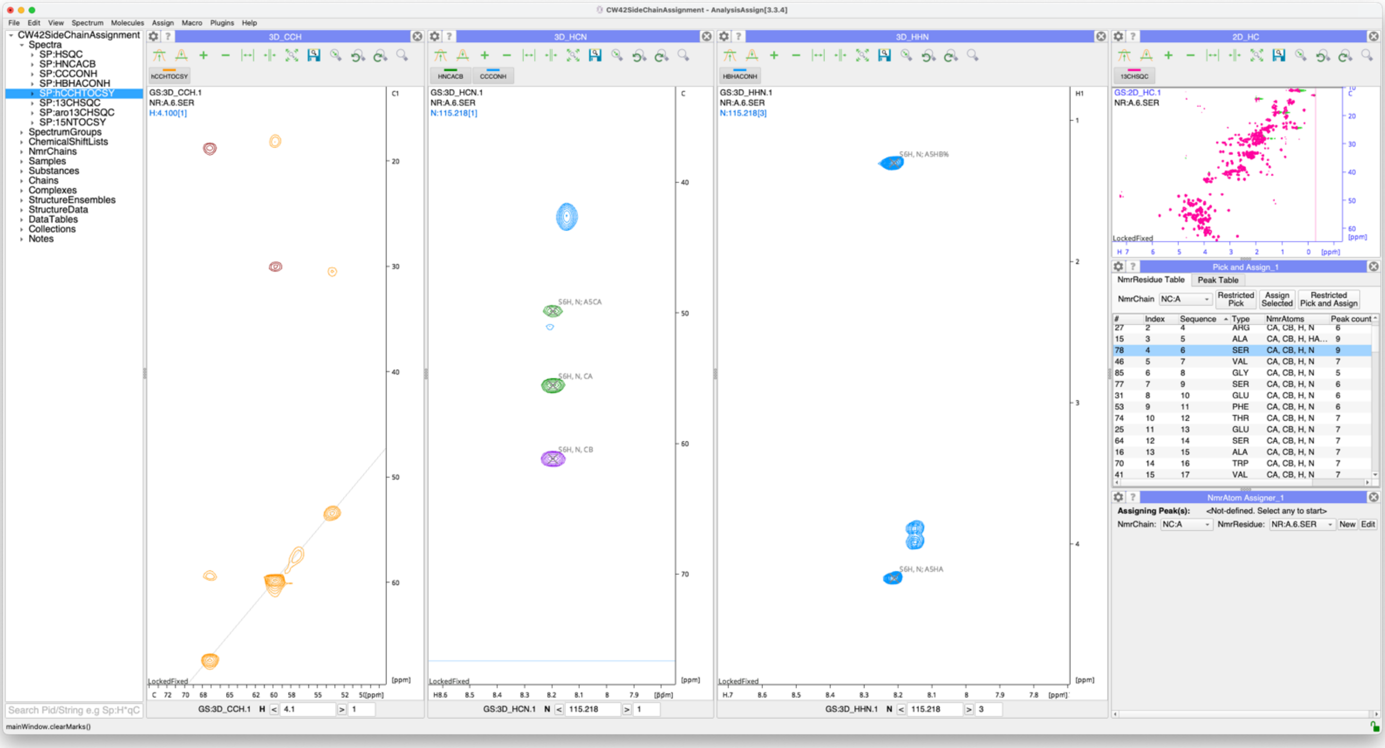
Histidine (HIS)

Note that you will need to select the i+1 residue in the **Pick and Assign** module to assign the HA/HB resonances. E.g. above, Ser 6 is selected, in order to assign the HA and HB% atoms in Ala 5.

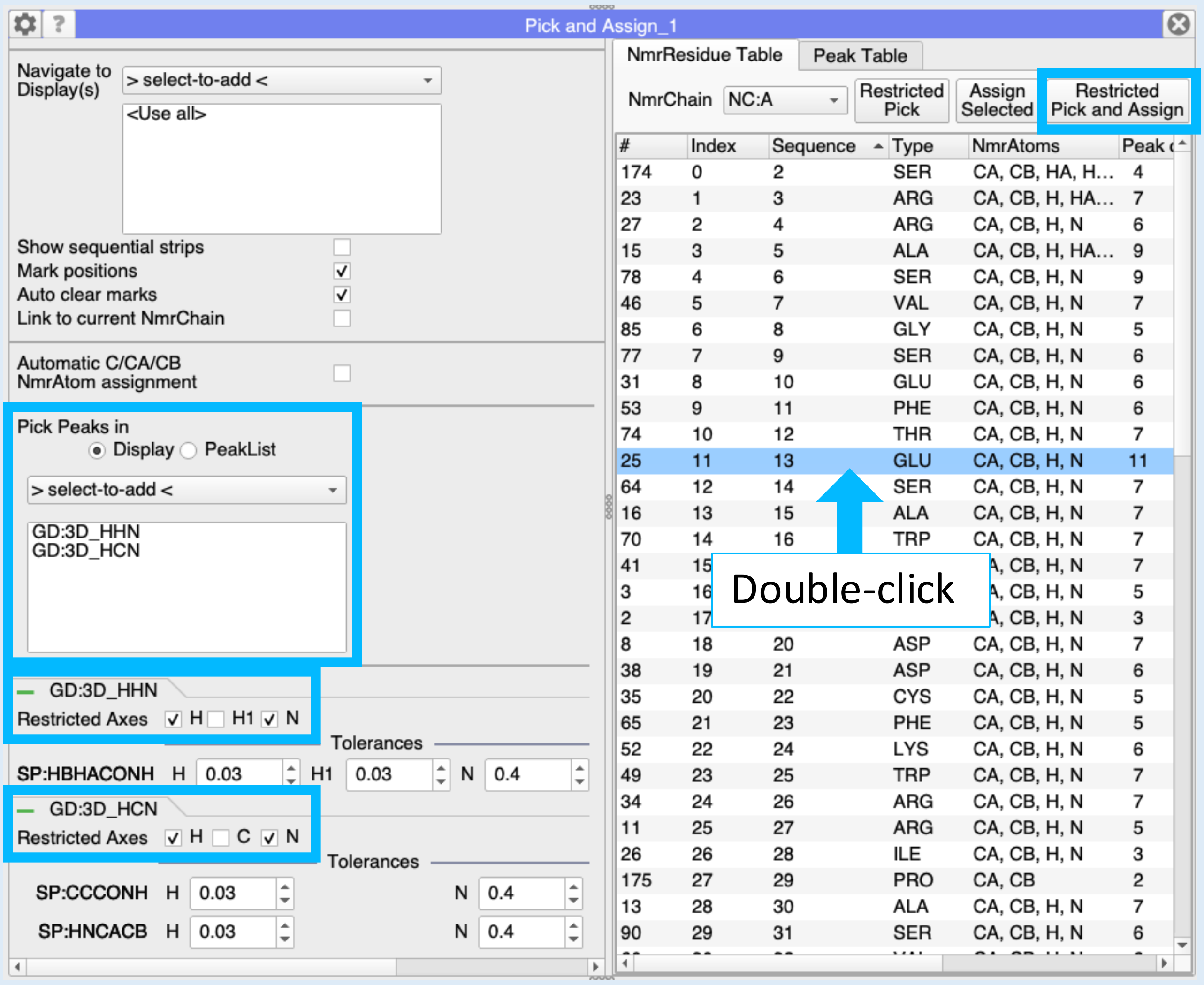


3c Adding additional spectra

- Close your **2D_HN** SpectrumDisplay using the X in the top right-hand corner
- **Drag** the **HNCACB** and **CCCONH** spectra from the sidebar into the drop area so as to open them in a new SpectrumDisplay.
- **Drag** the **hCCHTOCSY** spectrum into a new SpectrumDisplay.
- **Drag** the **13CHSQC** spectrum into a new SpectrumDisplay so that your modules are arranged roughly like this:



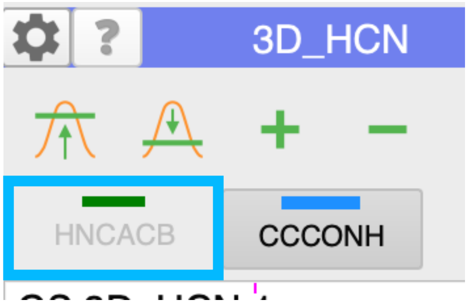
Assigning Long Side Chains

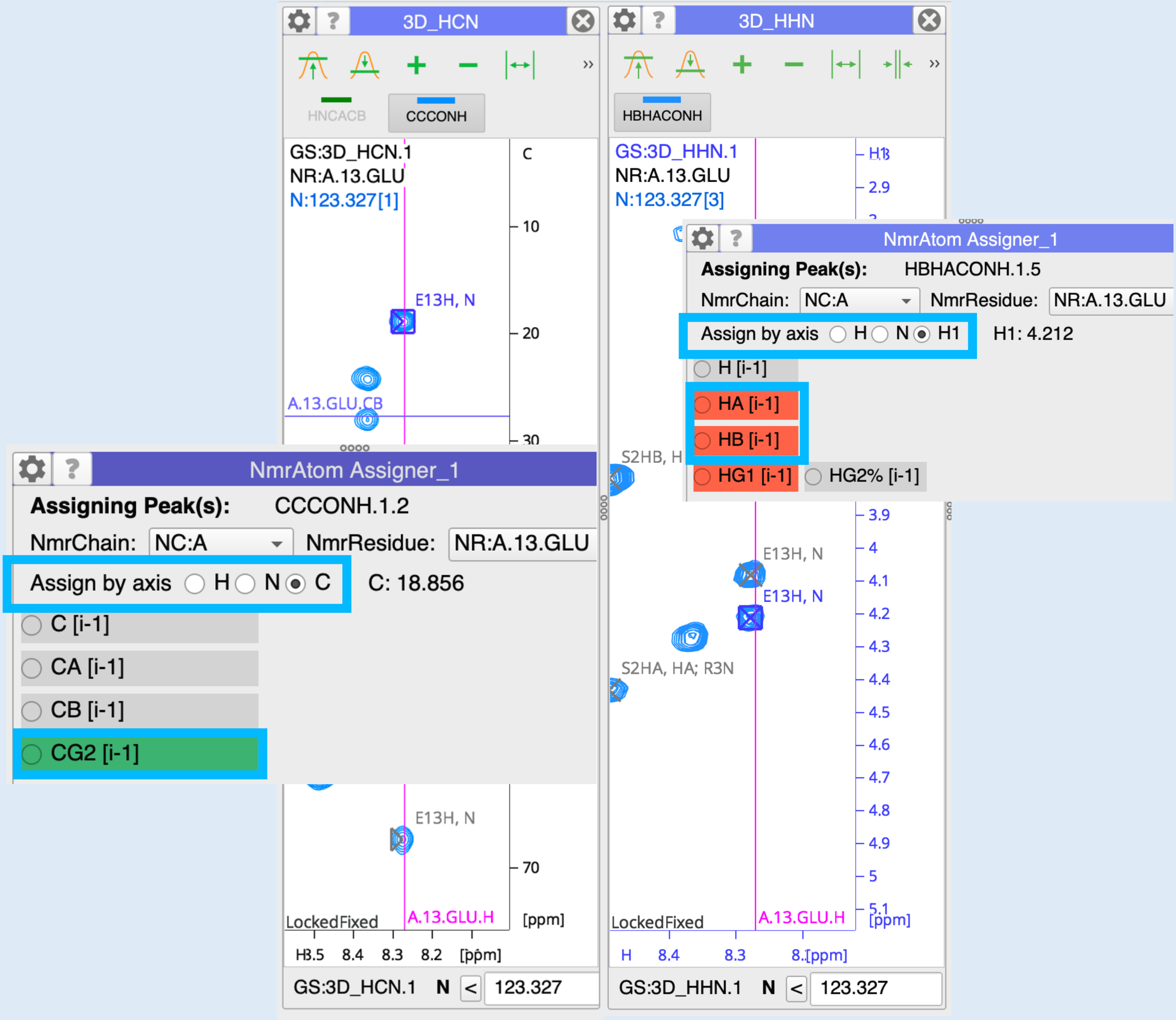


#	Index	Sequence	Type	NmrAtoms	Peak
174	0	2	SER	CA, CB, HA, H...	4
23	1	3	ARG	CA, CB, H, HA...	7
27	2	4	ARG	CA, CB, H, N	6
15	3	5	ALA	CA, CB, H, HA...	9
78	4	6	SER	CA, CB, H, N	9
46	5	7	VAL	CA, CB, H, N	7
85	6	8	GLY	CA, CB, H, N	5
77	7	9	SER	CA, CB, H, N	6
31	8	10	GLU	CA, CB, H, N	6
53	9	11	PHE	CA, CB, H, N	6
74	10	12	THR	CA, CB, H, N	7
25	11	13	GLU	CA, CB, H, N	11
64	12	14	SER	CA, CB, H, N	7
16	13	15	ALA	CA, CB, H, N	7
70	14	16	TRP	CA, CB, H, N	7
41	15			A, CB, H, N	7
3	16			A, CB, H, N	5
2	17			A, CB, H, N	3
8	18	20	ASP	CA, CB, H, N	7
38	19	21	ASP	CA, CB, H, N	6
35	20	22	CYS	CA, CB, H, N	5
65	21	23	PHE	CA, CB, H, N	5
52	22	24	LYS	CA, CB, H, N	6
49	23	25	TRP	CA, CB, H, N	7
34	24	26	ARG	CA, CB, H, N	7
11	25	27	ARG	CA, CB, H, N	5
26	26	28	ILE	CA, CB, H, N	3
175	27	29	PRO	CA, CB	2
13	28	30	ALA	CA, CB, H, N	7
90	29	31	SER	CA, CB, H, N	6

3B Pick and Assign HBHACONH / CCCONH peaks

- Open the **Pick and Assign** module **Settings**.
- For **Pick Peaks in Display** add the **GD:3D_HCN** SpectrumDisplay, so that you now have both the **GD:3D_HHN** and the **GD:3D_HCN** Spectrum Displays selected.
- Make sure that the **Restricted Axes** are set only to **H** and **N** for all spectra.
- **Double-click** on **13 GLU** in the NmrResidues table of the **Pick and Assign** module. The SpectrumDisplays will now navigate to the GLU 13 NH location. The **CCCONH** spectrum should show the THR 12 CA and CB peaks (overlapped with these same peaks in the HNCACB) as well as a new CG peak (at ~19 ppm).
- Switch the HNCACB spectrum off using its button in the Spectrum Toolbar.
- Make sure the **HBHACONH** and **CCCONH** spectra are at a suitable contour level for peak picking.
- Click on the **Restricted Pick and Assign** button in the **Pick and Assign** module.





3B Assigning HBHACONH / CCCONH side-chain NmrAtoms

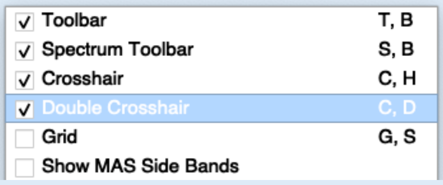
- Assign the HA and HB peaks for THR 12 using the **NmrAtom Assigner**, as in **Section 2**.

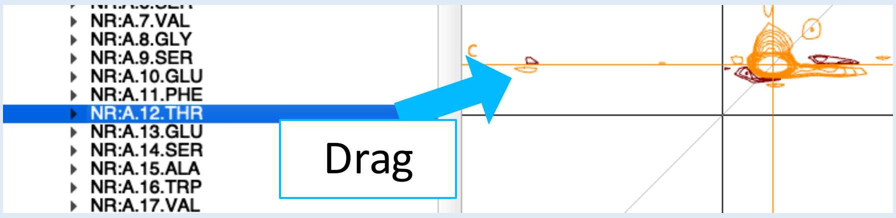
Assign the CG carbon in the **NmrAtom Assigner:**

- Select the **C** for Assign by Axis
- Select **CG2**.

3 Assigning Long Side Chains

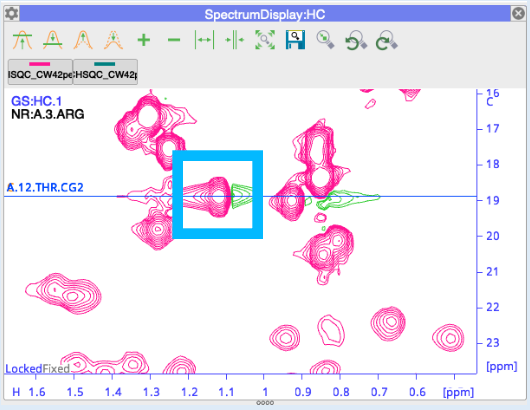
CW42SideChainAssignment

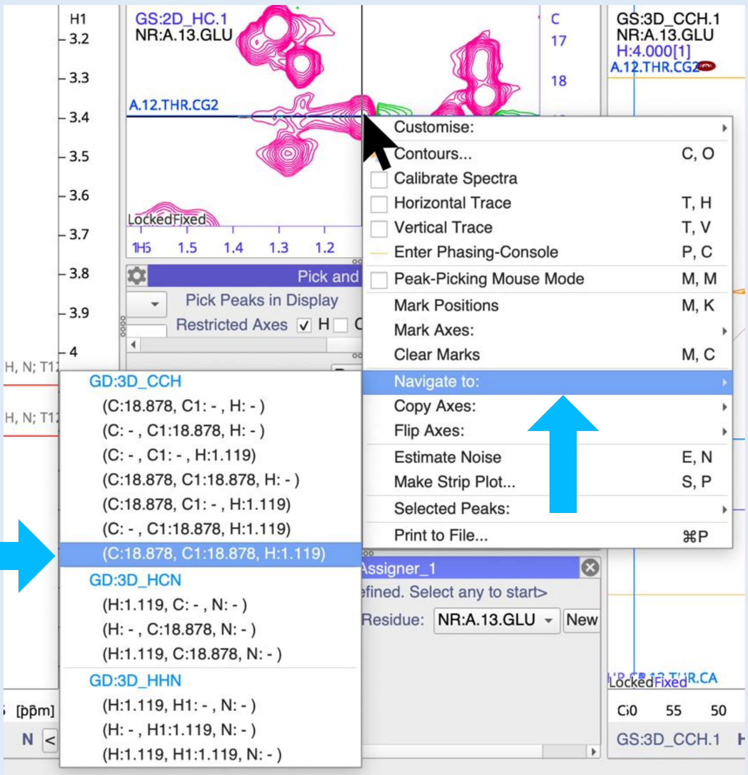




3D Placing marks

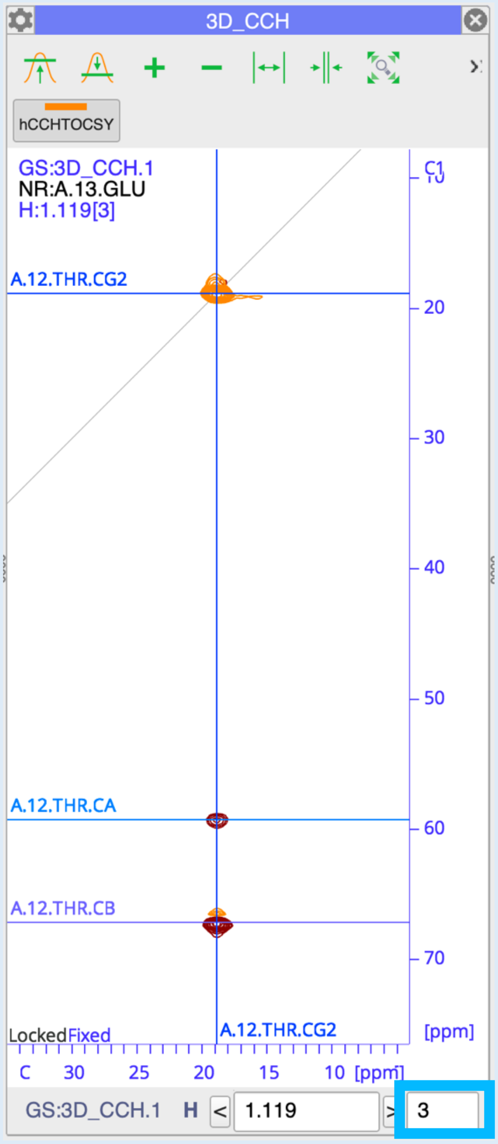
- **Right-click** in a Spectrum Display
- Clear all marks with **MC**.
- Drag the **NR:A.12.THR** NmrResidue from the sidebar into a Spectrum Display to mark all other Thr12 chemical shifts.

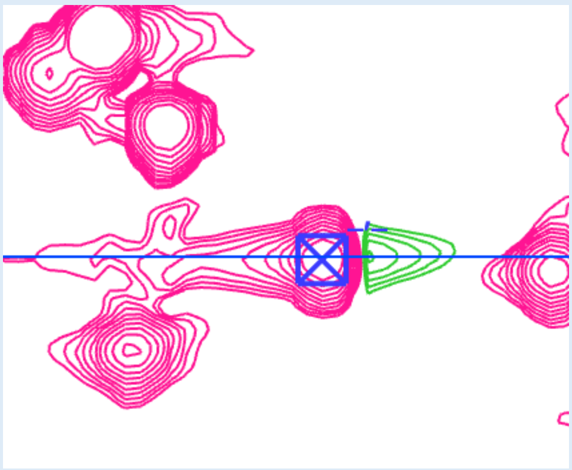


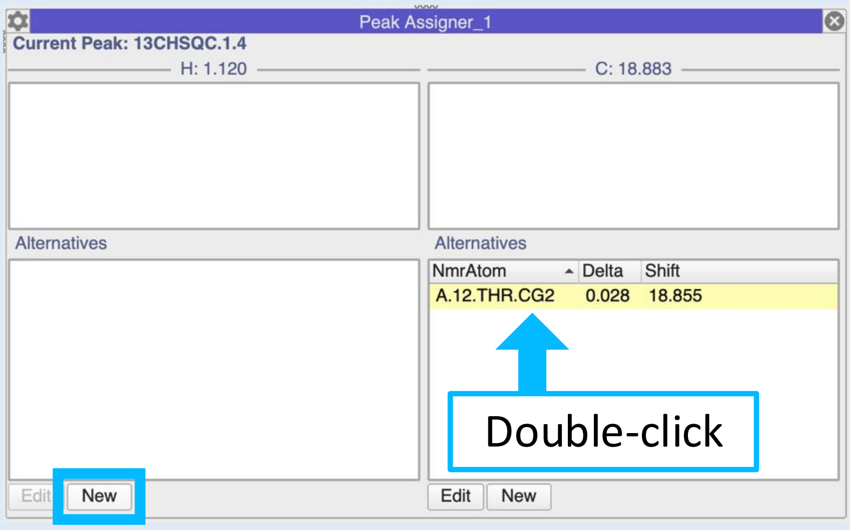


3E Navigating to the 13C-HSQC and hCCH-TOCSY

- In the **13CHSQC** spectrum, find the peak which the **A.12.THR.CG2** mark goes through.
- **Right-click** on this peak and go to **Navigate to:** and then in the **GD:3D_CCH** section select the final options which will navigate to all three dimensions.
  The **hCCH-TOCSY** spectrum in your **CCH** SpectrumDisplay will now navigate to this position.







3F Picking and assigning peaks

- Set the visible plan count in the **3D_CCH** SpectrumDisplay to 3.

The peaks at the intersection points of the Thr12 marks in the **hCCHTOCSY** spectrum confirm that these peaks belong to the Thr12 spin system and that the HG2% chemical shift is 1.12 ppm.

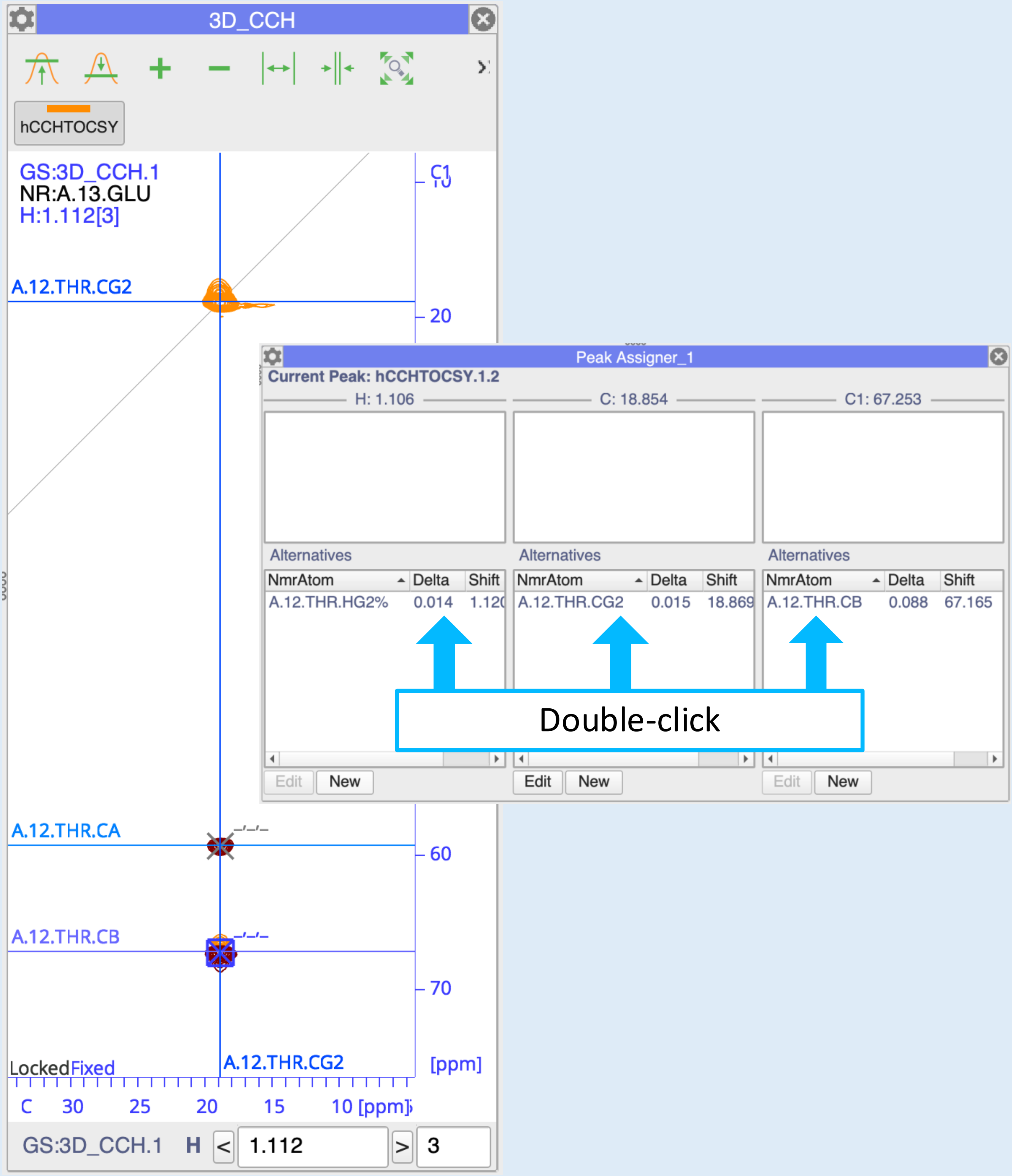
- Start by picking the CG2 peak in the **13CHSQC** spectrum.
- Open the **Peak Assigner** module by going to **Main Menu ⟶ Assign ⟶ Peak Assigner** or using the shortcut **AP**.

The left and right hand columns are for the H and C dimensions, respectively; the upper and lower section for the assignments and assignment options.

- **Double-click** on **A.12.THR.CG2** in the lower right section, to assign it to the C dimension of this peak. It will move to the upper right section.

For the H dimension we still need to the create the Thr 12 HG2% NmrAtom:

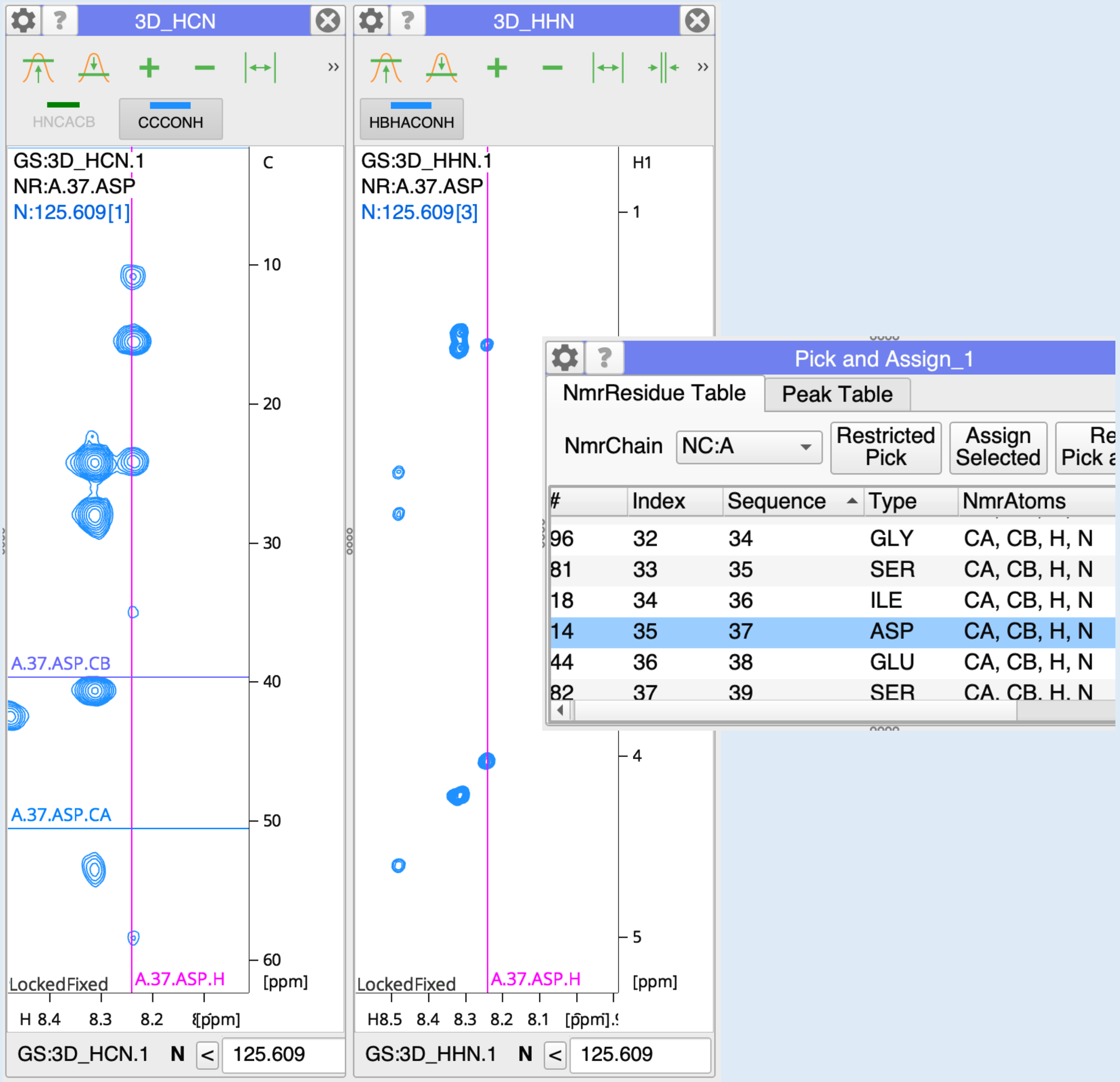
- Click on New on the left hand side.
- Fill in the fields as shown and click **Accept**.



3G Picking and Assigning peaks contd.

- Now pick the two non-diagonal peaks in the **hCCHTOCSY** spectrum (at the CA and CB chemical shifts).
- In the **Peak Assigner**, assign all three dimensions of both peaks by **double-clicking** the correct alternatives on the right hand side.

If you ever accidentally select an incorrect option from the **Alternatives** below (e.g. if more than one is available), simply **double-click** on it in the upper box and it will move back down to the **Alternatives** again.



3H Assigning further long side chains

Try assigining the side-chains of

Ile36

Glu60

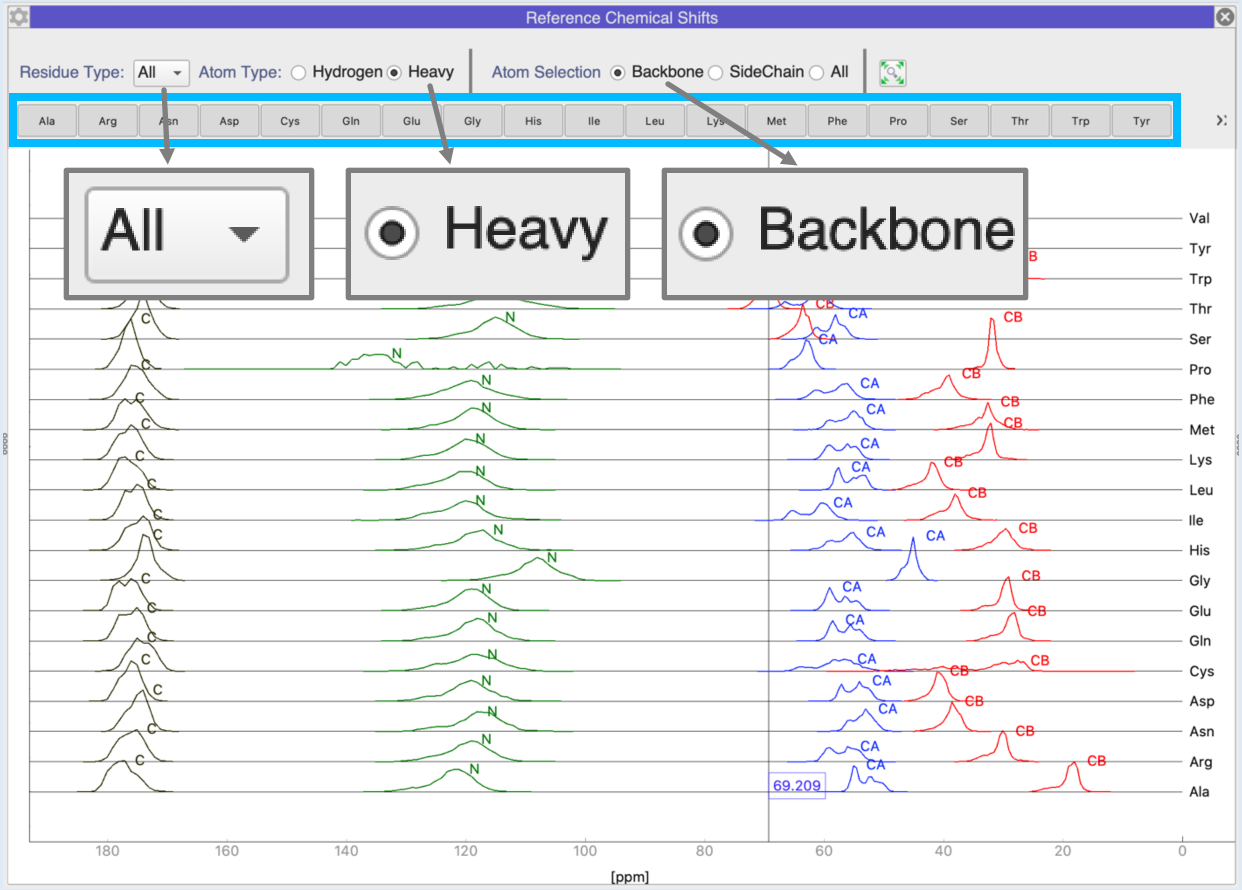
Pro29

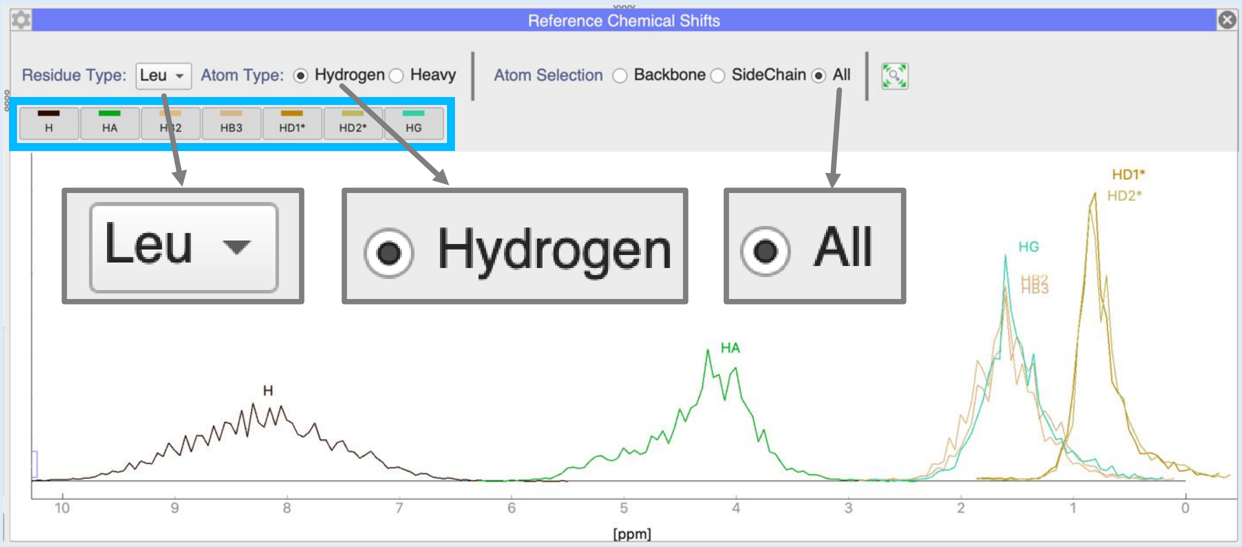
Gln74

Leu70

You may wish to refer to some of the reference material at the start of the tutorial, e.g. the typical carbon chemical shifts for the 20 natural amino acids and their IUPAC/NEF atom names.

The next page shows how you can access reference chemical shifs from within the program.





RC

3| Reference Chemical Shifts

You can check the standard chemical shifts for protein amino acids within CcpNmr Analysis:

- Go to **Main Menu ⟶ Molecules ⟶ Reference Chemical Shifts**, or type **RC**.
- For the **Residue Type** select either **All** or an individual amino acid, e.g. **Leu**.
- For the and **Atom Type** select either **Hydrogen** or **Heavy**.
- For the **Atom Selection** select **Backbone**, **SideChain** or **All**.
- Switch off particular amino acid or atom types in the toolbar.

A mouse cursor correlates the ppm position with that in your SpectrumDisplays. You can move the graph or zoom with the mouse wheel on axes or in the main graph area like in a SpectrumDisplay.

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Cite Us

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